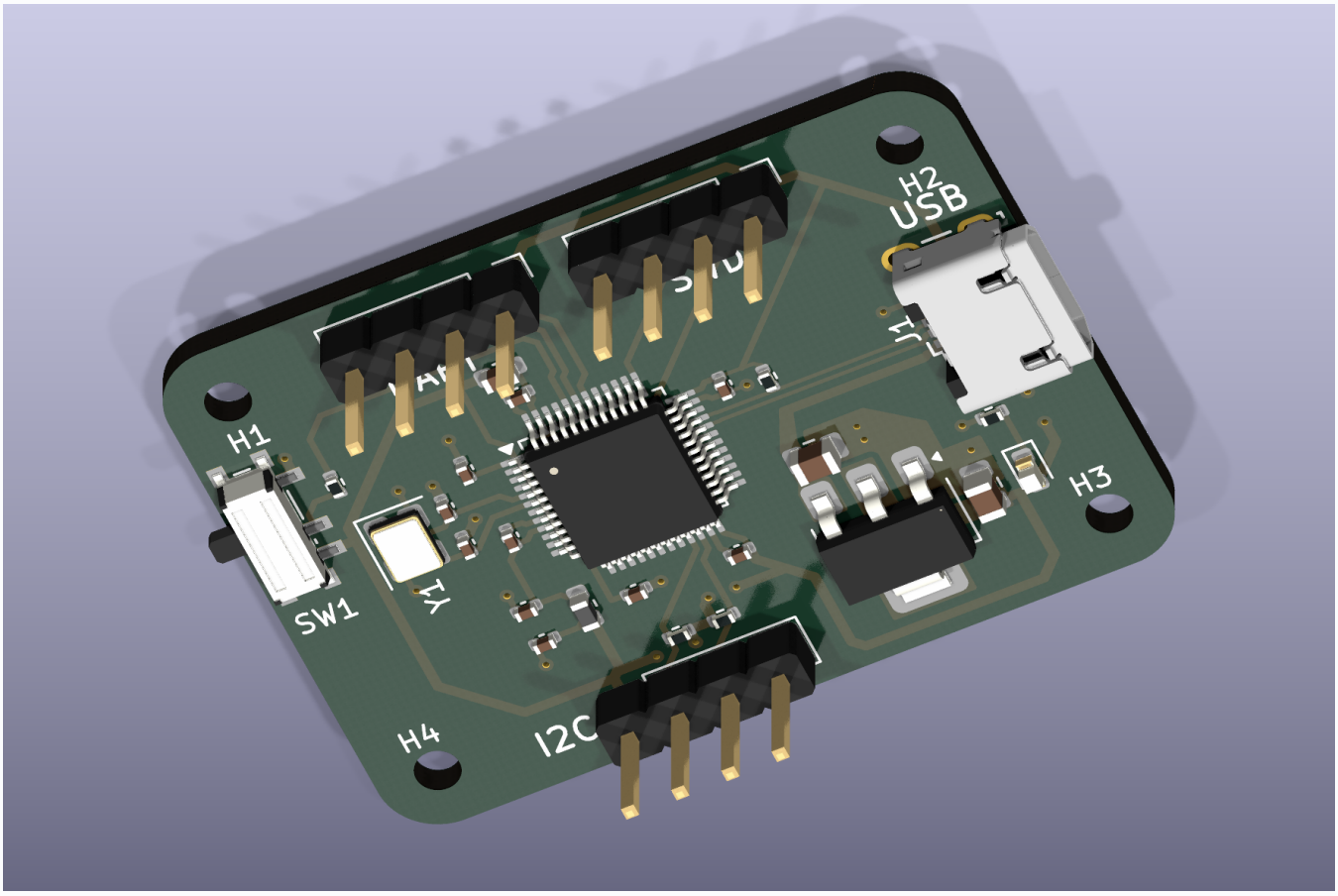
HARDWARE DESIGN PORTFOLIO

STM32 I2C

Development Board

USB-powered hardware for 3.3 V sensor interfaces

Stavros Stratakis | Project brief | 11 October 2026



KiCad 3D rendering of the documented board; not a prototype photograph.

40 × 28.75
mm outline

2 layers
1.6 mm FR-4

3.3 V
on-board rail

16 MHz
crystal

A compact STM32F103 board combining power regulation, I2C sensor connectivity, UART diagnostics and SWD programming. The design package includes editable CAD, circuit documentation and manufacturing exports.

Design status Routed and checked in KiCad. Physical assembly, firmware and bench validation remain the next stage.

01 / FUNCTION

How the board works

Power, processing and peripheral access on one small PCB.

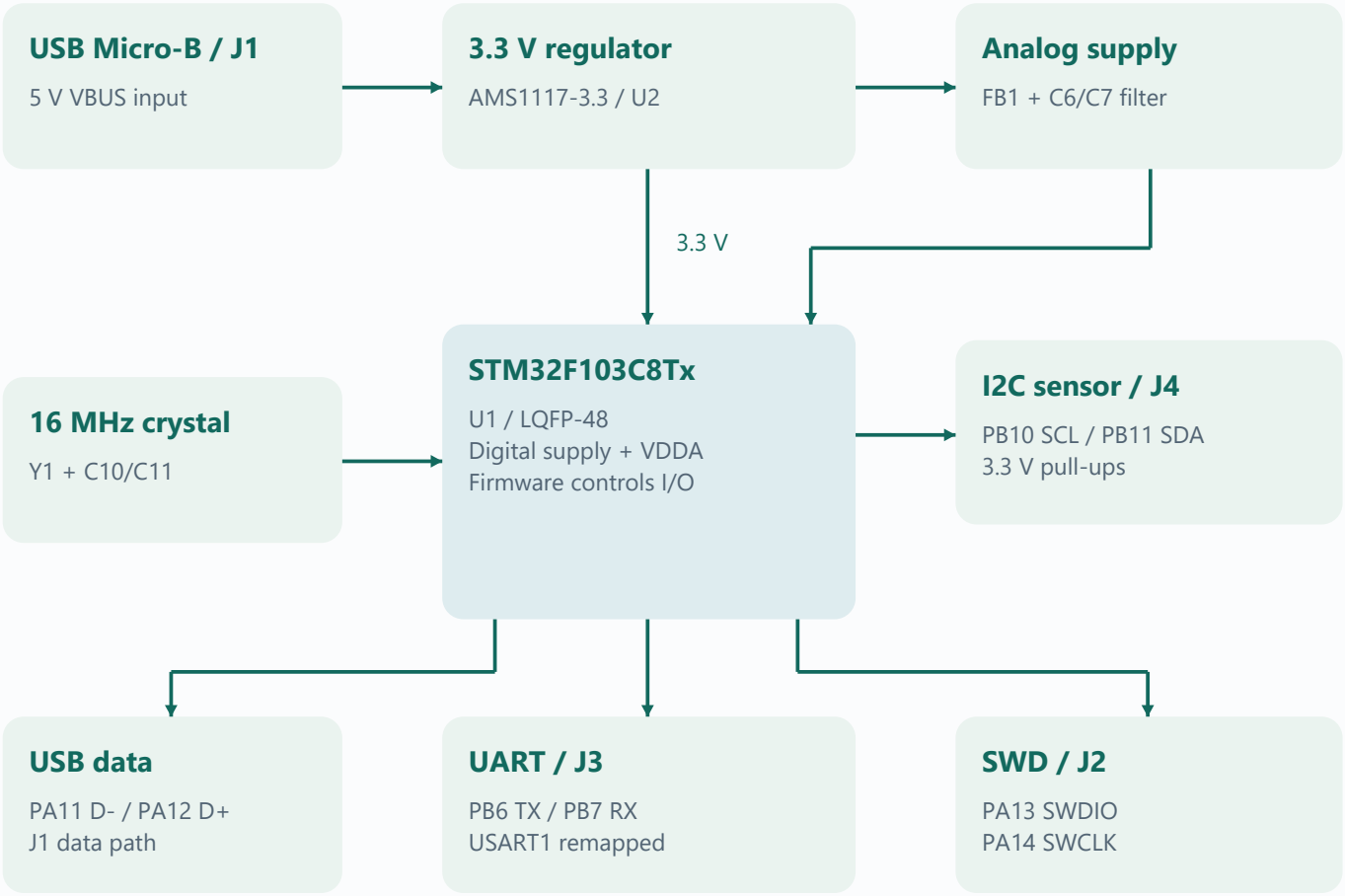


Diagram shows logical connections; signal direction depends on the interface.

Intended operating sequence

1. Power. USB VBUS feeds the regulator. The 3.3 V rail powers the MCU and sensor header; the LED indicates that the rail is present.

2. Program. Use SWD to load firmware, configure the clocks and select the required peripherals.

3. Acquire and report. Firmware can read a connected I2C sensor and send results over UART or a USB device implementation.

02 / CIRCUIT DECISIONS

Support circuitry

The components around the MCU establish power, clocking and bus behavior.

Power regulation and decoupling

U2 converts nominal 5 V USB power to 3.3 V. C12 and C13 are 22 µF; C1-C4 provide local 100 nF bypassing and C5 adds 10 µF bulk capacitance. FB1 and its capacitors filter the analog supply.

Clock and reset

Y1 is a 16 MHz crystal with two 10 pF load capacitors. C9 is 100 nF on NRST. SW1 selects BOOT0 through a 10 kΩ resistor; normal flash execution uses BOOT0 low.

Two useful design calculations

Regulator: (5.0 - 3.3) V × 0.10 A = **0.17 W** dissipation, excluding quiescent current.
I2C: 3.3 V / 1.5 kΩ ≈ **2.2 mA** sink current per low line, before accounting for VOL.

Calculated examples, not measured ratings or test results.

I2C and serial access

J4 exposes I2C2 on PB10/PB11. R4 and R5 each pull the bus to 3.3 V through 1.5 kΩ. J3 exposes USART1 on PB6/PB7; firmware must enable the remapped pin assignment.

USB and programming

J1 routes USB D+/D- directly to the MCU, with a fixed 1.5 kΩ D+ pull-up. USB data operation needs firmware and a valid 48 MHz USB clock. J2 provides a four-pin SWD programming interface.

Checks before building a prototype

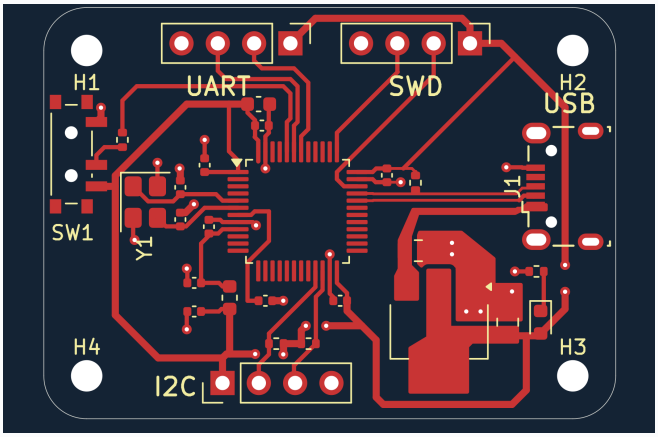
Match the exact regulator and output capacitor for stability; select the crystal by load capacitance and ESR; check sensor pull-ups and cable capacitance. Confirm purchased component pinouts against the saved footprints.

Circuit details and references →

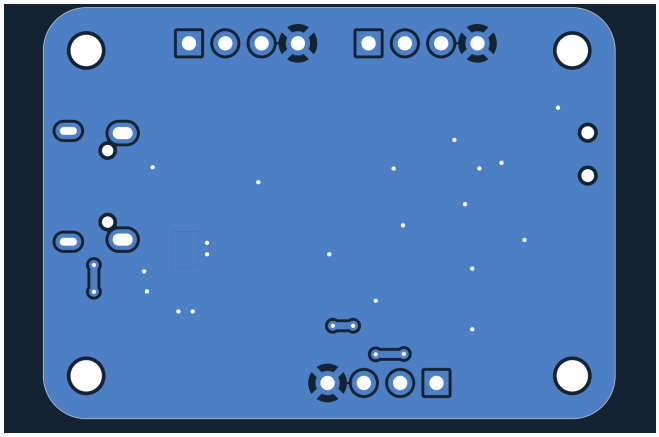
03 / IMPLEMENTATION

PCB layout and interfaces

A two-layer board with accessible edge connectors and a central MCU.



Top copper + silkscreen



Bottom copper, viewed from below

The crystal sits near the oscillator pins. USB and regulation occupy the right edge; headers sit along the top and bottom. The lower copper layer is predominantly ground, with short signal routes and stitching vias.

Header pinout

Pin	J4 / I2C	J3 / UART	J2 / SWD
1	+3.3 V	+3.3 V	+3.3 V reference
2	SCL / PB10	TX / PB6	SWDIO / PA13
3	SDA / PB11	RX / PB7	SWCLK / PA14
4	GND	GND	GND

Orientation: pin 1 is the square PCB pad. Use the assembly map to identify J2, J3 and J4. SWD pin 1 is directly on the 3.3 V rail; avoid driving it from another enabled supply.

Interface limits: there is no 5 V I2C level translator and no NRST pin on the SWD header. Use 3.3 V peripherals and a 3.3 V logic UART adapter.

04 / EVIDENCE

Verification and deliverables

CAD checks completed with KiCad 9.0.6 on 11 October 2026.

0	0	0	0
ERC violations	DRC violations	Unconnected items	Parity issues

Results apply to the committed rule settings and libraries. No exclusions were added. Fabricator-specific edge and mask constraints still require review; these reports do not establish electrical performance.

Included in the package

Editable KiCad project and local footprints; schematic PDF; board renders and layer views; design BOM; Gerbers and plated/unplated drills; SMD placement; assembly map; verification reports and source checksums.

Next physical milestones

Complete part selection and fabrication review. Assemble the prototype, measure the power rails, establish SWD access, then validate the clock, UART, I2C transactions and USB operation with firmware.

Current revision boundary

The routed hardware design and review package are available. There is no assembled-board test evidence or STM32 firmware in this revision. USB ESD protection and a verified impedance specification are not included.

Explore the evidence

Schematic PDF

Design and pinout

Manufacturing package

Verification reports

Bring-up procedure

Library and model provenance

Design source: repository revision **6f40200**. All PCB figures are generated from the saved KiCad design. Library symbols, footprints and manufacturer models retain their respective provenance.